Worksheet Week 9 #2 – Objects
EC327 Introduction to Software Engineering – Spring 2026

1. What purpose do classes serve?
A. data encapsulation
B. providing a convenient way of modeling real-world objects
C. simplifying code reuse
D. all of the above

2. Which is not a protection level provided by classes in C++?
A. protected
B. hidden
C. private
D. public

3. What value must a destructor return?
A. A pointer to the class.
B. An object of the class.
C. A status code determining whether the class was destructed correctly
D. Destructors do not return a value.

4. Which of the following is a valid class declaration?
A. class A { int x; };
B. class B { }
C. public class A { }
D. object A { int x; };

5. Which functions will every class contain?
A. None
B. Constructor
C. Destructor
D. Both a constructor and a destructor

6. What does your class can hold?
a) data
b) functions
c) both a & b
d) none of the mentioned

7. How many specifiers are present in access specifiers in class?
a) 1
b) 2
c) 3
d) 4

8. Which is used to define the member of a class externally?
a) :
b) ::
c) #
d) none of the mentioned

9. Which other keywords are also used to declare the class other than class?
a) struct
b) union
c) object
d) both a & b

10. What is the output of this program?

```
#include <iostream>
using namespace std;
class rect
{
    int x, y;
    public:
    void val (int, int);
    int area ()
    {
          return (x * y);
      }
  };
  void rect::val (int a, int b)
  {
      x = a;
      y = b;
  }
  int main ()
  {
      rect rect;
      rect.val (3, 4);
      cout << "rect area: " << rect.area();
      return 0;
  }
```

a) rect area:12
b) rect area: 12
c) rect area:24
d) none of the mentioned

11. What is the output of this program?

```
#include <iostream>
using namespace std;
class CDummy
{
    public:
    int isitme (CDummy& param);
};
int CDummy::isitme (CDummy& param)
{
      if (&param == this)
          return true;
      else
          return false;
  }
  int main ()
  {
      CDummy a;
      CDummy *b = &a;
      if (b->isitme(a)) {
          cout << "execute";
      }
      else
      {
          cout<<"not execute";
      }
      return 0;
  }
```

a) execute
b) not execute
c) none of the mentioned
d) both a & b

12. The fields in the class in c++ program are by default
a) protected
b) private
c) public
d) none of the mentioned

13. Constructors are used to
a) initalize the objects
b) construct the data members
c) both a & b
d) none of the mentioned

14. When struct is used instead of the keyword class means, what will happen in the program?
a) access is public by default
b) access is private by default
c) access is protected by default
d) none of the mentioned

15. An int i is a static data member of class X if Please select the best answer.

A. it is declared static within the class
B. it is shared by all variables of that class
C. it is accessible as X::i
D. all of the above

16. The this pointer: Please select the best answer.

A. is const and therefore cannot have its value altered
B. is available for native types
C. is mutable and can have its value altered
D. must be explicitly declared in a class

17. A static member function: Please select the best answer.

A. does not have access to this pointer arguments
B. must be public
C. is a class variable
C. none of the above

18. If class IN is declared in class OUT and int i is in IN, how would you reference i unambiguously? Please select the best answer.

A. IN::OUT::i
B. OUT::i
C. OUT::IN::i
D. ::i